Algebraic Reasoning & Properties — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: Every step needs a reason*

Key Idea

In geometry we don’t just get the answer — we **justify** every step with a **reason**: a definition, a property, a postulate or a theorem.

☞ *Video: Solve and justify*

Watch Me — Worked Example

$x - 4 = 10$ Given

$x = 14$ **Addition Property of Equality**

The reason names **what you did** to both sides.

☞ *Video: Properties of equality*

Key Idea

Addition / Subtraction	add or subtract the **same** number on both sides
Multiplication / Division	multiply or divide both sides by the same number (not 0)
Substitution	if $a = b$, you can **replace** a with b
Distributive	$a(b + c) =$ $ab + ac$

☞ *Video: Reflexive, symmetric, transitive*

Key Idea

Reflexive $a = a$ e.g. $AB = AB$

Symmetric if $a = b$, then $b = a$

Transitive if $a = b$ and $b = c$, then $a = c$

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☞ *Video: A longer example*

Watch Me — Worked Example

$3(x+2)=21$	Given
$3x+6=21$	**Distributive Property**
$3x=15$	**Subtraction Property of Equality**
$x=5$	**Division Property of Equality**

☞ *Video: Your turn*

Your Turn

Name the property.

(a) If $AB=CD$, then $CD=AB$. **Symmetric** — the equation is flipped.

(b) If $m\angle 1=m\angle 2$ and $m\angle 2=m\angle 3$, then $m\angle 1=m\angle 3$. **Transitive** — a chain.

Notice

Remember: Every step needs a **reason**. Did something to both sides? Name that property of equality. **Reflexive** $a=a$, **symmetric** flip, **transitive** chain.

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